Sets

Q1. Subsets. Decide whether each statement is true or false and prove that your decision is correct.

(a) $\{x \in \mathbb{R} \mid \tan(x) = 1\} \subseteq \{x \in \mathbb{R} \mid \sin(x) \leq \cos(x)\}$

(b) $\left\{\cos\left(\frac{k\pi}{3}\right) \,\middle|\, k \in \mathbb{Z}\right\} \subseteq \mathbb{Z}$

(c) $\left\{\cos\left(\frac{k\pi}{3}\right) \,\middle|\, k \in \mathbb{Z}\right\} \subseteq \mathbb{Q}$

Q2. Unions and complements. Express each of the following sets as a union of intervals and as a set complement. Drawing pictures may help.

(a) $\{x \in \mathbb{R} : 1 - x^2 < 3x^2\}$.

(b) $\{x \in \mathbb{R} : \sin(x) \geq 0\} \cap [0, 6\pi)$.

(c) $\{x \in \mathbb{R} : e^{-x} \geq e^x\}$.

(d) $\{x \in [0, 2\pi] \mid \sin(x) < \cos(x)\}$.

Q3. More Set Calculations. Express the following sets in the simplest possible form.

(a) $\mathbb{Q} \cap \{x \in \mathbb{R} : 3^x = 2\}$

(b) $\mathbb{Q} \cap \{x \in \mathbb{R} : 2x^2 - 3x + 1 = 0\}$

(c) $\mathbb{Z} \cap \{x \in \mathbb{R} : 2x^2 - 3x + 1 = 0\}$.

(d) $\{x \in \mathbb{R} : \sin(x) \in \mathbb{Z}\}$.

Q4. Solving inequalities. Solve the following inequalities, expressing your answers in set notation.

(a) $3 - 2x \leq -1 - x$

(b) $3 - 2x \leq -1 - 2x$

(c) $1 \leq \frac{1}{x^2} < 4$

Q5. More inequalities Solve the following inequalities, expressing your answers in set notation. You may use the fact that log is order preserving on its entire domain. That is, if $x < y$, then $\log(x) < \log(y)$

(a) $e^{2x} - 2 \geq e^x$ [Hint: The quadratic formula will be useful.]

(b) $1 - \cos^2(x) \leq \sin^2(x)$

(c) $1 - \cos^2(x) < \sin^2(x)$

Complex Numbers

Note: In these problems, the principal argument should be used whenever the complex number is written in polar or exponential form, that is $-\pi < \theta \leq \pi$.

Q6. Square root of negative numbers. Represent the following square roots as imaginary numbers.

(a) $\sqrt{-4}$

(b) $\sqrt{-6}$

(c) $2\sqrt{-12}$

Q7. Real and Imaginary. In each case, find $\mathrm{Re}(z)$, $\mathrm{Im}(z)$, and $\mathrm{Re}(z)$–$\mathrm{Im}(z)$.

(a) $z = 3 - 4i$

(b) $z = 6i$

(c) $z = 2 + 7i$

(d) $z = \frac{5}{7}$ (e) $z = \frac{1}{2}i + 2$

Complex Arithmetic

Q8. Complex Arithmetic. Let $z_1 = 2+6i,\ z_2 = \frac{1}{2}+2i,\ z_3 = -6i,\ z_4 = -3+2i$ and $z_5 = 5$. Simplify:

(a) $z_1 + z_4$ (b) $z_1 + z_3$ (c) $z_1 - 2z_2$

(d) $z_4 + z_5$ (e) $z_2 + z_4$ (f) $3z_2 - 2z_4$

(g) $z_5 + z_3$ (h) $z_3 + z_2$ (i) $2z_1 + z_2 + 3z_3$

Q9. The Complex Plane. Let $z_1 = 2+4i,\ z_2 = -2+4i,\ z_3 = 2-4i$ and $z_4 = -2-4i$.

(a) Sketch z_1, z_2, z_3 and z_4 on the same set of axes. (b) Sketch $z_1, 2z_1$ and $-3z_1$ on the same set of axes. What do you observe?

(c) Sketch $z_1 + z_2,\ z_1 + z_4$ and $z_2 + z_3$ on the same set of axes.

(d) Is it true that

(i) $z_1 = -z_2$? (ii) $z_1 = -z_3$?

(iii) $z_1 = -z_4$? (iv) $z_2 = -z_3$?

Q10. Powers of i. Express i^2, i^3 and i^4 in Cartesian form (without using polar form). Use your answers to express the following in cartesian form:

(a) $i^{23} = (i^4)^5 \times i^3$ (b) i^{14} (c) i^{259}

Q11. Cartesian form. Simplify each of the following products. You should express your answer in the form $a + ib,\ a, b \in \mathbb{R}$.

(a) $i(2+3i)$ (b) $(3-2i)^2$ (c) $(-2-i)(2+i)$

(d) $3(2+3i)$ (e) $(2+i)(2-i)$ (f) $(1+i)^3$

(g) $(3-2i)(1+i)$ (h) $(2+i)(-2+i)$ (i) $i^2(3+2i)$

Q12. Rationalising the Denominator. Write each of the following expressions $\frac{z_1}{z_2}$ in the form $a + ib,\ a, b \in \mathbb{R}$. In each case you should first write down $\overline{z_2}$.

x (a) $\dfrac{1}{3-2i}$ (b) $\dfrac{2}{2+i}$ (c) $\dfrac{i}{2+3i}$ (d) $\dfrac{5-2i}{1-i}$

x (e) $\dfrac{4+3i}{-2+i}$ (f) $\dfrac{1-i}{-3-2i}$

Q13. Real and Imaginary. Find the real and imaginary parts:

(a) $\text{Im}\left(\dfrac{1-5i}{4+i}\right)$ (b) $\text{Re}\left(\dfrac{2+7i}{4-6i}\right)$

Q14. Simplifying Complex Numbers. Express each of the following complex numbers in the form

$a + ib,\ a, b \in \mathbb{R}$.

(a) $\dfrac{i(1+2i)}{3(2-i)}$ (b) $\dfrac{3(2-i)}{i(1+2i)}$

(c) $\dfrac{(2+i)^2}{(1+3i)(2+3i)}$ (d) $\dfrac{(3+2i)(5-i)}{i^2(2-i)}$

Q15. Conjugates to Cartesian. Express the following numbers in the cartesian form $a + ib$ with a, b real:

(a) $\overline{12+7i}$ (b) $(1+i)^2 + (\overline{1+i})$ (c) $(\overline{3+2i})(4+5i)$

(d) $(\overline{3-i})(2-i)(-1-3i)$ (e) $\overline{i^2(1-i)^3}$ (f) $\overline{\left(\dfrac{2-3i}{1+5i}\right)}$

Q16. Complex conjugates. Let $z = x + iy$ with x and y real, and let $\bar{z}$ be the complex conjugate. Check the following identities:

(a) $z\bar{z} = x^2 + y^2$ (b) $z + \bar{z} = 2\,\mathrm{Re}(z)$ (c) $z - \bar{z} = 2i\,\mathrm{Im}(z)$

Sketch the following points in the complex plane:

(d) z and $\bar{z}$ (e) z and $-z$ (f) z and $\overline{-z}$

Modulus and Argument

Q17. Modulus and Argument. For each of the following complex numbers write down $|z|$ (the modulus of z) and Arg(z) (remember this is an angle in the range $(-\pi, \pi]$). You may find it useful to sketch the complex numbers on an argand diagram first.

(a) $z = 6$ (b) $z = -3$ (c) $z = -\sqrt{3} + i$

(d) $z = 4i$ (e) $z = -2i$ (f) $z = -1 - \sqrt{3}i$

(g) $z = 7i$ (h) $z = 1 + i$ (i) $z = 4 - 4i$

Q18. Something more complicated.

(a) Find the modulus of each of the following complex numbers without multiplying them into cartesian form:

(i) $\dfrac{5+2i}{2+5i}$ (ii) $\dfrac{-27i(8+2i)(2+i)}{(4+i)(4-3i)(4-8i)}$ (iii) $\dfrac{(2-3i)(-2-2i)(-5)}{(1+i)(3+2i)}$

(b) Find the principal argument $\mathrm{Arg}(z)$ for each of the following complex numbers:

(i) $\dfrac{-2}{1+i\sqrt{3}}$ (ii) $\dfrac{i}{-2-2i}$ (iii) $(1+i)^4$

Regions in the Complex Plane

Q19. Regions in the Complex Plane. Sketch the following subsets in the complex plane:

(a) $\{z \in \mathbb{C} : \mathrm{Re}(z) = 1\}$ (b) $\{z \in \mathbb{C} : \mathrm{Im}(z) > 1\}$

(c) $\{z \in \mathbb{C} : |z - 2i| \leq 1\}$ (d) $\{z \in \mathbb{C} : |z - 3| = |z + 2i|\}$

(e) $\{z \in \mathbb{C} : |z| \leq 4, \mathrm{Arg}(z) \in [0, \pi/4]\}$

Complex Exponential

Q20. Exponential Polar Form. Express each of the following complex numbers in polar form $re^{i\theta}$. In each case, chose an angle θ, $-\pi < \theta \leq \pi$:

(a) -1 (b) $5 + 5i$

(c) $(1-i)^2$ (d) $2 + 5i$

Q21. Powers. Use exponential polar form to simplify the following powers of complex numbers:

(a) $(1+i)^{20}$ (b) $(2\sqrt{3} + 2i)^5$

(c) $\left(\dfrac{1+i}{\sqrt{3}+i}\right)^{12}$ (d) $\left(\dfrac{1-i}{\sqrt{3}-i}\right)^{7}$

Finding Complex Solutions

Q22. Complete the square. Find all solutions for z in each of the following equations, completing the square where necessary.

(a) $z^2 + 1 = 0$ (b) $z^2 + 16 = 0$ (c) $z^2 - 4 = 0$

(d) $z^2 + 4z + 5 = 0$ (e) $z^2 + 3z + 3 = 0$

Q23. Quadratic formula. Find all solutions for z in each of the following equations by using the quadratic formula.

(a) $z^2 + 3z + 2 = 0$ (b) $z^2 + 3z + 4 = 0$

(c) $2z^2 - z + 3 = 0$ (d) $3z^3 + 2z^2 + z = 0$

Q24. Any way you like.

Find all solutions for z in each of the following equations. These can be done by factoring, or by rearranging and finding the nth root.

(a) $z^4 - 81 = 0$

(b) $z^8 - 256 = 0$.

Q25. Roots and Factors. For each of the following find the roots of the given equations and sketch the roots in the complex plane. Express your answers using principal arguments:

(a) cube roots $z^3 = 1$

(b) square roots $z^2 = i$

(c) sixth roots $z^6 = -64$

(d) fifth roots $z^5 = 32e^{5\pi i/3}$

Q26. Roots of Polynomials. Find the roots of the following polynomials, using the complex exponential and roots of unity where necessary. Express your answers as sets.

(a) $z^4 + 4z^2 + 4 = 0$

(b) $z^4 + 4z^2 + 16 = 0$

(c*) $(z + i)^5 - (z - i)^5 = 0$

(d) $z^4 + z^3 + z^2 + z + 1 = 0$

Topic 1&2: Sets and proofs, complex numbers

A1. Subsets.

(a) True.
Proof: Let $x \in \{x \in \mathbb{R} \mid \tan(x) = 1\}$, so

$$\begin{aligned}\tan(x) = 1 &\Rightarrow \frac{\sin(x)}{\cos(x)} = 1 \\ &\Rightarrow \sin(x) = \cos(x) \\ &\Rightarrow \sin(x) \leq \cos(x) \\ &\Rightarrow x \in \{x \in \mathbb{R} \mid \sin(x) \leq \cos(x)\}\end{aligned}$$

(b) False.
Example: For $k = 1$ we have $\cos\left(\frac{k\pi}{3}\right) = \frac{1}{2} \notin \mathbb{Z}$.

(c) True.
Proof: Let $x \in \left\{\cos\left(\frac{k\pi}{3}\right) \,\middle|\, k \in \mathbb{Z}\right\}$, so $x = \cos\left(\frac{k\pi}{3}\right)$ for some $k \in \mathbb{Z}$. A unit circle diagram shows $x \in \{-1, -\frac{1}{2}, \frac{1}{2}, 1\}$, so $x \in \mathbb{Q}$.

A2. Unions and complements.

(a) $(-\infty, -\frac{1}{2}) \cup (\frac{1}{2}, \infty) = \mathbb{R} \setminus [-\frac{1}{2}, \frac{1}{2}]$

(b) $[0, \pi] \cup [2\pi, 3\pi] \cup [4\pi, 5\pi] = [0, 5\pi] \setminus ((\pi, 2\pi) \cup (3\pi, 4\pi))$

(c) $(-\infty, 0]$.

(d) $\left[0, \frac{\pi}{4}\right) \cup \left(\frac{5\pi}{4}, 2\pi\right] = [0, 2\pi] \setminus \left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$.

A3. More Set Calculations.

(a) $\varnothing$ (b) $\{\frac{1}{2}, 1\}$. (c) $\{1\}$.

(d) $\{k\pi \mid k \in \mathbb{Z}\} \cup \{k\pi + \frac{\pi}{2} \mid k \in \mathbb{Z}\} = \{\frac{k\pi}{2} \mid k \in \mathbb{Z}\}$.

A4. Solving inequalities.

(a) $x \in [4, \infty)$ (b) $x \in \varnothing$ (c) $x \in [-1, -\frac{1}{2}) \cup (\frac{1}{2}, 1]$

A5. More inequalities

(a) $x \in [\log(2), \infty)$ (b) $\mathbb{R}$ (c) $\varnothing$

A6. Square root of negative numbers.

(a) $2i$ (b) $\sqrt{6}i$ (c) $4\sqrt{3}i$.

A7. Real and Imaginary.

(a) $\mathsf{Re}(z) = 3$, $\mathsf{Im}(z) = -4$, $\mathsf{Re}(z) - \mathsf{Im}(z) = 7$

(b) $\mathsf{Re}(z) = 0$, $\mathsf{Im}(z) = 6$, $\mathsf{Re}(z) - \mathsf{Im}(z) = -6$

(c) $\mathsf{Re}(z) = 2$, $\mathsf{Im}(z) = 7$, $\mathsf{Re}(z) - \mathsf{Im}(z) = -5$

(d) $\mathsf{Re}(z) = \frac{5}{7}$, $\mathsf{Im}(z) = 0$, $\mathsf{Re}(z) - \mathsf{Im}(z) = \frac{5}{7}$

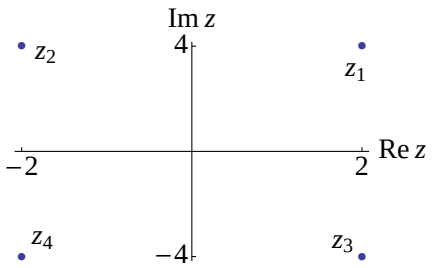
(e) $\mathsf{Re}(z) = 2$, $\mathsf{Im}(z) = \frac{1}{2}$, $\mathsf{Re}(z) - \mathsf{Im}(z) = \frac{3}{2}$.

A8. Complex Arithmetic.

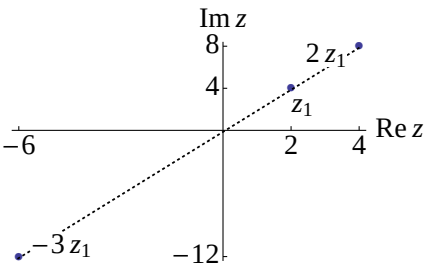
(a) $-1+8i$ (b) 2 (c) $1+2i$ (d) $2+2i$ (e) $-\frac{5}{2}+4i$ (f) $\frac{15}{2}+2i$ (g) $5-6i$ (h) $\frac{1}{2}-4i$ (i) $\frac{9}{2}-4i$.

A9. The Complex Plane.

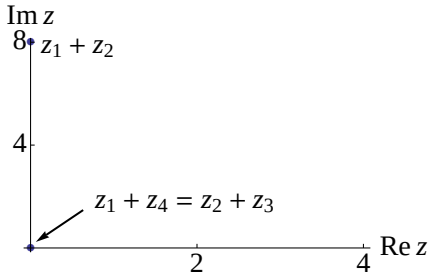
(a)



(b)



(c)



All points are on the same line.

(d) (i) No (ii) No (iii) Yes (iv) Yes.

A10. Powers of i.

$i^2 = -1$, $i^3 = -i$, $i^4 = 1$ (a) $-i$ (b) -1 (c) $-i$

A11. Cartesian form.

(a) $-3+2i$ (b) $5-12i$ (c) $-3-4i$ (d) $6+9i$ (e) 5 (f) $-2+2i$ (g) $5+i$ (h) -5 (i) $-3-2i$.

A12. Rationalising the Denominator.

(a) $\frac{3}{13}+\frac{2}{13}i$ (b) $\frac{4}{5}-\frac{2}{5}i$ (c) $\frac{3}{13}+\frac{2}{13}i$ (d) $\frac{7}{2}+\frac{3}{2}i$ (e) $-1-2i$ (f) $-\frac{1}{13}+\frac{5}{13}i$.

A13. Real and Imaginary. (a) -21/17 (b) -17/26

A14. Simplifying Complex Numbers.

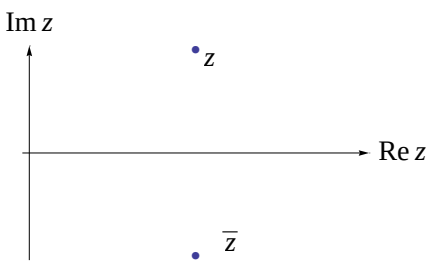
(a) $-\frac{1}{3}$ (b) -3 (c) $\frac{3}{26}-\frac{11}{26}i$ (d) $-\frac{27}{5}-\frac{31}{5}i$.

A15. Conjugates to Cartesian.

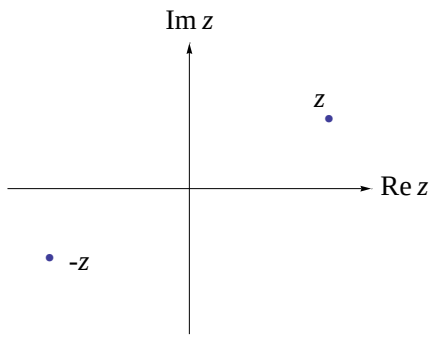
(a) $12-7i$ (b) $1+i$ (c) $22+7i$ (d) $-10-20i$ (e) $2-2i$ (f) $-\frac{1}{2}+\frac{1}{2}i$.

A16. Complex conjugates.

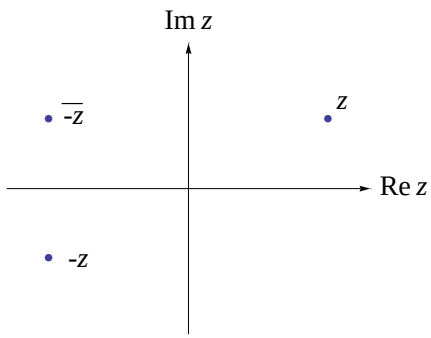
(d) reflection in x axis



(e) reflection in origin



(f) reflection in y axis.



A17. Modulus and Argument.

(a) $|z| = 6$, Arg$(z) = 0$ (b) $|z| = 3$, Arg$(z) = \pi$ (c) $|z| = 2$, Arg$(z) = \frac{5\pi}{6}$

(d) $|z| = 4$, Arg$(z) = \frac{\pi}{2}$ (e) $|z| = 2$, Arg$(z) = -\frac{\pi}{2}$ (f) $|z| = 2$, Arg$(z) = -\frac{2\pi}{3}$

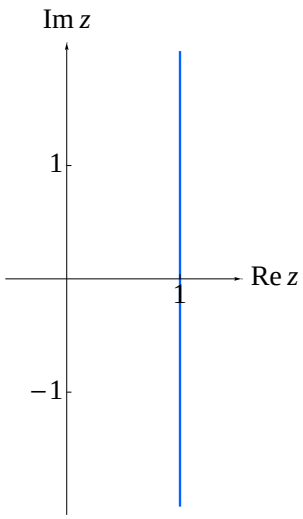
(g) $|z| = 7$, $\text{Arg}(z) = \frac{\pi}{2}$ (h) $|z| = \sqrt{2}$,$\text{Arg}(z) = \frac{\pi}{4}$ (i) $|z| = 4\sqrt{2}$, $\text{Arg}(z) = -\frac{\pi}{4}$.

A18. Something more complicated.

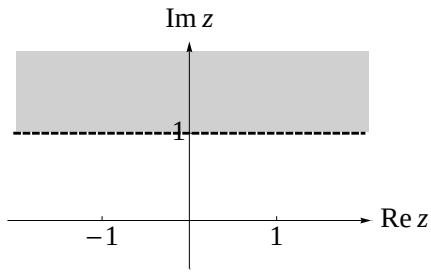
(a) (i) 1 (ii) 27/10 (iii) 10 (b) (i) $2\pi/3$ (ii) $-3\pi/4$ (iii) π

A19. Regions in the Complex Plane.

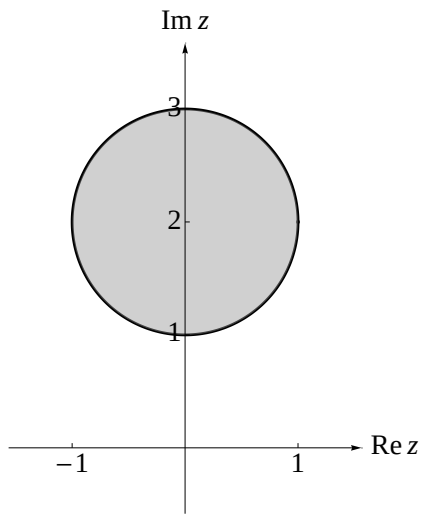
(a) Line $x = 1$



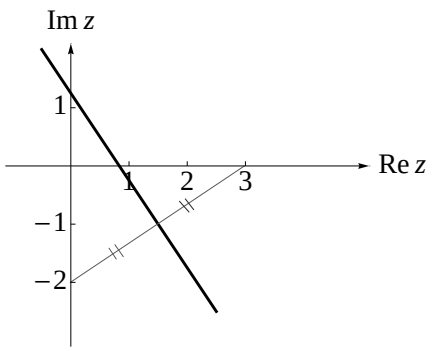
(b) half plane $y > 1$



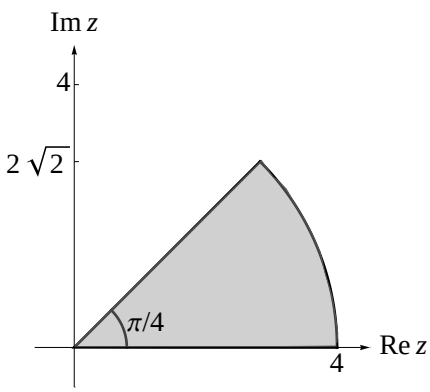
(c) boundary and interior of the circle $x^2 + (y-2)^2 = 1$



(d) line $6x + 4y = 5$



(e) Sector of circle, centred at the origin, radius 4, angle $\pi/4$ anticlockwise from the real axis.



A20. Exponential Polar Form.

(a) $e^{i\pi}$ (b) $5\sqrt{2}e^{i\pi/4}$ (c) $2e^{-i\pi/2}$ (d) $\sqrt{29}e^{i\alpha}$, where $\alpha = \arctan(5/2)$

A21. Powers. (a) -1024 (b) $-512\sqrt{3} + 512i$ (c) $-1/64$ (d) $\frac{1}{2^{7/2}}e^{-7\pi i/12}$

A22. Complete the square.

(a) $z_1 = i, \quad z_2 = -i$ (b) $z_1 = 4i, \quad z_2 = -4i$ (c) $z_1 = 2, \quad z_2 = -2$ (d) $z_1 = -2 + i, \quad z_2 = -2 - i$
(e) $z_1 = -\frac{3}{2} + \frac{\sqrt{3}}{2}i, \quad z_2 = -\frac{3}{2} - \frac{\sqrt{3}}{2}i.$

A23. Quadratic formula.

(a) $z_1 = -2, \quad z_2 = -1$ (b) $z_1 = -\frac{3}{2} + \frac{\sqrt{7}}{2}i, \quad z_2 = -\frac{3}{2} - \frac{\sqrt{7}}{2}i$ (c) $z_1 = \frac{1}{4} + \frac{\sqrt{23}}{4}i, \quad z_2 = \frac{1}{4} - \frac{\sqrt{23}}{4}i$

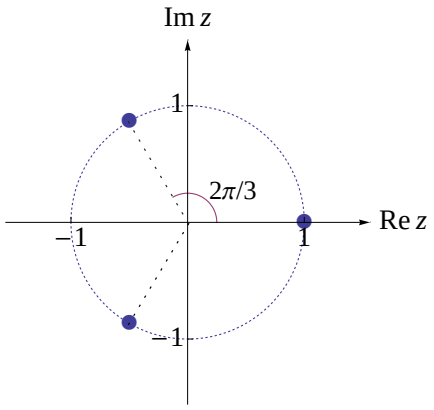
(d) $z_1 = -\frac{1}{3} + \frac{\sqrt{2}}{3}i, \quad z_2 = -\frac{1}{3} - \frac{\sqrt{2}}{3}i, z_3 = 0$

A24. Any way you like.

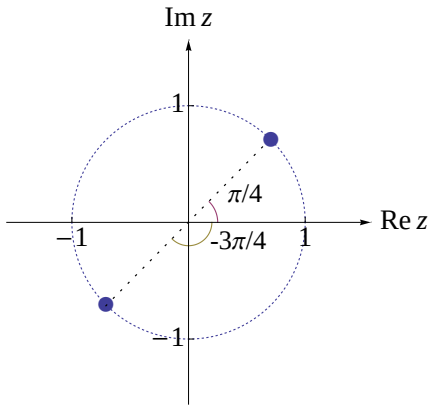
(a) $z_1 = 3, \quad z_2 = -3, \quad z_3 = 3i, \quad z_4 = -3i$
(b) $z_1 = 2, \quad z_2 = -2, \quad z_3 = 2i, \quad z_4 = -2i, \quad z_5 = -\sqrt{2} - \sqrt{2}i, \quad z_6 = \sqrt{2} + \sqrt{2}i, \quad z_7 = \sqrt{2} - \sqrt{2}i,$
$z_8 = -\sqrt{2} + \sqrt{2}i.$

A25. Roots and Factors.

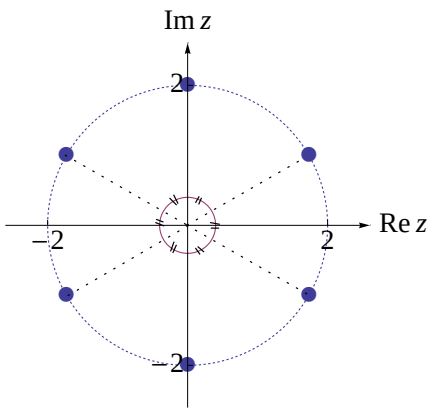
(a) 1, $e^{2\pi i/3}$, $e^{-2\pi i/3}$



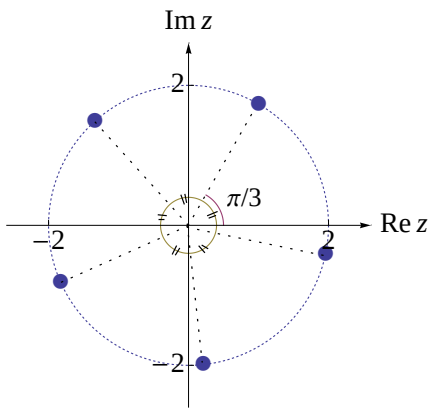
(b) $e^{\pi i/4}$, $e^{-3\pi i/4}$



(c) $2e^{\pi i/6}$, $2e^{\pi i/2}$, $2e^{5\pi i/6}$, $2e^{-5\pi i/6}$, $2e^{-\pi i/2}$, $2e^{-\pi i/6}$



(d) $2e^{\pi i/3}$, $2e^{11\pi i/15}$, $2e^{-13\pi i/15}$, $2e^{-7\pi i/15}$, $2e^{-\pi i/15}$



A26. Roots of Polynomials.

(a) $\{-i\sqrt{2}, i\sqrt{2}\}$ (b) $\{-1 - i\sqrt{3}, -1 + i\sqrt{3}, 1 - i\sqrt{3}, 1 + i\sqrt{3}\}$

(c) $\left\{-\sqrt{(5+2\sqrt{5})/5}, \sqrt{(5+2\sqrt{5})/5}, \sqrt{(5-2\sqrt{5})/5}, -\sqrt{(5-2\sqrt{5})/5}\right\}$

(d) $\{e^{2\pi i/5}, e^{4\pi i/5}, e^{6\pi i/5}, e^{8\pi i/5}\}$.

Useful Formulae

Pythagorean identity

$$\cos^2(x) + \sin^2(x) = 1$$

Compound angle formulae

$$\begin{aligned}
\sin(x+y) &= \sin(x)\cos(y) + \cos(x)\sin(y) \\
\cos(x+y) &= \cos(x)\cos(y) - \sin(x)\sin(y) \\
\tan(x+y) &= \frac{\tan(x)+\tan(y)}{1-\tan(x)\tan(y)}
\end{aligned}$$

Derivatives of inverse trigonometric functions

$$\begin{aligned}
\arcsin'(x) &= \frac{1}{\sqrt{1-x^2}} \\
\arccos'(x) &= -\frac{1}{\sqrt{1-x^2}} \\
\arctan'(x) &= \frac{1}{1+x^2}
\end{aligned}$$

Antiderivatives from inverse trigonometric functions

$$\begin{aligned}
\int \frac{1}{\sqrt{s^2-x^2}}\,dx &= \arcsin\left(\frac{x}{s}\right) + C \\
\int \frac{-1}{\sqrt{s^2-x^2}}\,dx &= \arccos\left(\frac{x}{s}\right) + C \\
\int \frac{1}{s^2+x^2}\,dx &= \frac{1}{s}\arctan\left(\frac{x}{s}\right) + C
\end{aligned}$$

where s is a *positive* constant, and C is an arbitrary constant of integration.

Complex exponential formulae

$$\begin{aligned}
e^{i\theta} &= \cos(\theta) + i\sin(\theta) \\
\cos(\theta) &= \frac{1}{2}\left(e^{i\theta} + e^{-i\theta}\right) \\
\sin(\theta) &= \frac{1}{2i}\left(e^{i\theta} - e^{-i\theta}\right)
\end{aligned}$$

Vector projections

- The *vector projection* of $\mathbf{v}$ onto $\mathbf{u}$ is $\mathbf{v}_{\parallel} = (\hat{\mathbf{u}} \cdot \mathbf{v})\hat{\mathbf{u}} = k\mathbf{u}$, where $k \in \mathbb{R}$ is the unique solution of $\mathbf{u}\cdot(\mathbf{v}-k\mathbf{u}) = 0$.
- The *vector component of* $\mathbf{v}$ *perpendicular to* $\mathbf{u}$ is $\mathbf{v}_{\perp} = \mathbf{v} - \mathbf{v}_{\parallel}$.

Complex roots

The n-th roots of $w = se^{i\phi}$ are $s^{\frac{1}{n}}\, e^{i(\frac{1}{n}(\phi+2k\pi))}$ for $k = 0, 1, \ldots, n-1$.

Changes in speed

Provided $\mathbf{r}'(t) \neq \mathbf{0}$, the speed function $\|\mathbf{r}'(t)\|$ is decreasing when $\mathbf{r}'(t)\cdot\mathbf{r}''(t) < 0$ and increasing when $\mathbf{r}'(t)\cdot\mathbf{r}''(t) > 0$.